

1. Write a Pandas program to create a line plot of the historical stock prices of Alphabet Inc. between two specific dates.

Aim:

To create a line plot of the historical stock prices of Alphabet Inc. between two specific dates using Pandas and Matplotlib.

Algorithm

1. Import the required libraries (pandas and matplotlib.pyplot).
2. Read the stock price dataset (alphabet_stock.csv) using pd.read_csv().
3. Convert the Date column into datetime format using pd.to_datetime().
4. Specify the start date and end date for the required period.
5. Filter the dataset to include only the records between the selected dates.
6. Plot the Closing Price against the Date using plt.plot().
7. Add the chart title, x-axis label, y-axis label, and grid.
8. Display the line plot using plt.show().

Code:

```
import pandas as pd
import matplotlib.pyplot as plt # type: ignore

df = pd.read_csv("Csv/Lab4.csv")

df["Date"] = pd.to_datetime(df["Date"])
start_date = "2022-01-05"
end_date = "2022-01-12"

stock = df[(df["Date"] >= start_date) & (df["Date"] <= end_date)]

plt.figure(figsize=(8,5))
plt.plot(stock["Date"], stock["Close"], marker="o")
```

```
plt.title("Historical Stock Prices of Alphabet Inc.")
```

```
plt.xlabel("Date")
```

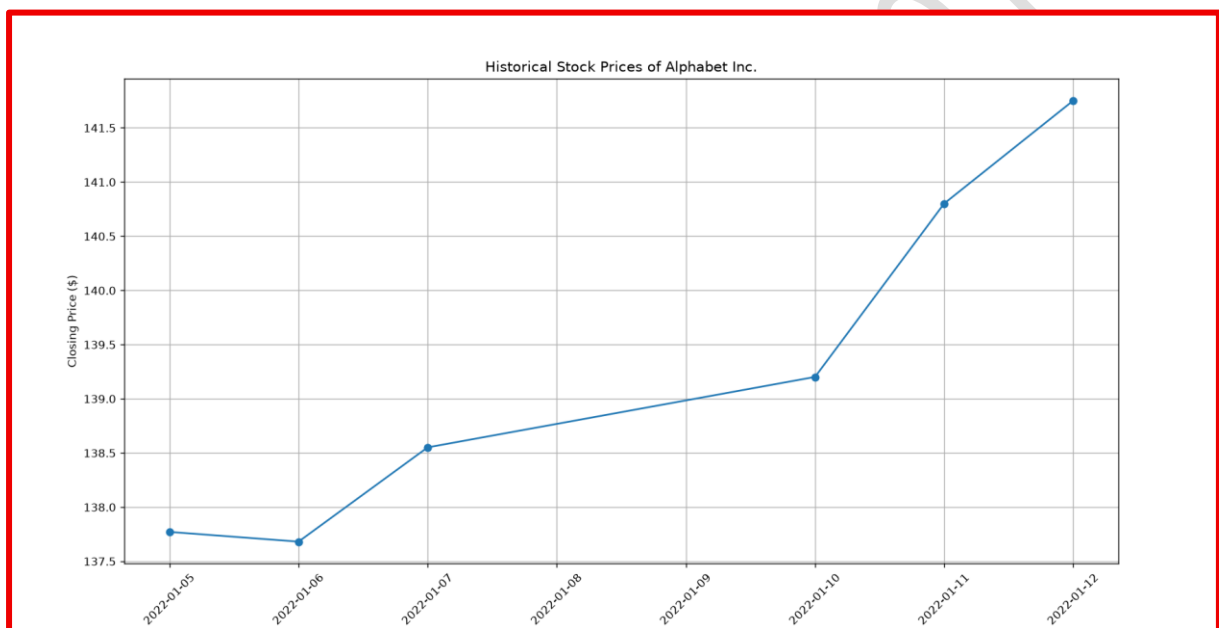
```
plt.ylabel("Closing Price ($)")
```

```
plt.grid(True)
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

Output:



Result:

The historical stock prices of Alphabet Inc. between the specified dates were successfully filtered and displayed as a line graph using Pandas and Matplotlib. The graph clearly shows the variation in the company's closing stock prices over the selected time period.

2. Write a Pandas program to create a bar plot of the trading volume of Alphabet Inc. stock between two specific dates.

Aim:

To create a bar plot of the trading volume of Alphabet Inc. stock between two specific dates using Pandas and Matplotlib.

Algorithm

1. Import the required libraries (pandas and matplotlib.pyplot).
2. Read the stock dataset (alphabet_stock.csv) using pd.read_csv().
3. Convert the Date column into datetime format using pd.to_datetime().
4. Specify the start date and end date.
5. Filter the records between the selected dates.
6. Plot the Volume against the Date using plt.bar().
7. Add the chart title, axis labels, and rotate the date labels.
8. Display the bar plot using plt.show()

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Csv/Lab4.csv")

df["Date"] = pd.to_datetime(df["Date"])

start_date = "2022-01-05"
end_date = "2022-01-12"

stock = df[(df["Date"] >= start_date) & (df["Date"] <= end_date)]

plt.figure(figsize=(8,5))
plt.bar(stock["Date"].dt.strftime("%Y-%m-%d"), stock["Volume"])
```

```
plt.title("Trading Volume of Alphabet Inc.")
```

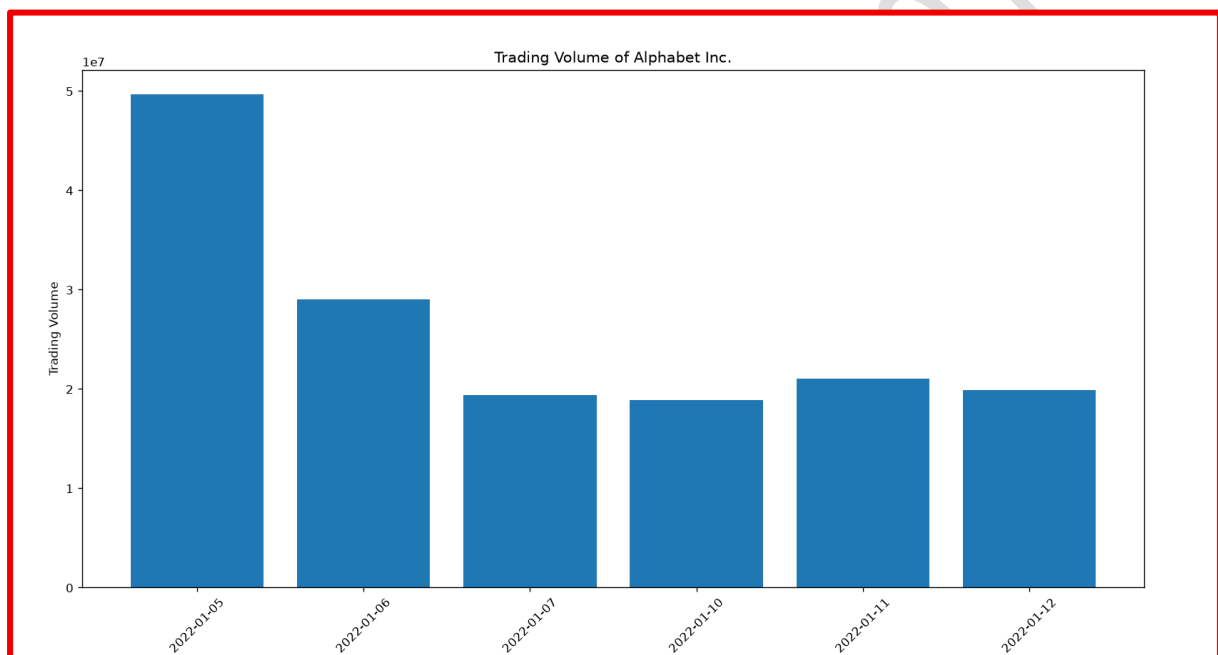
```
plt.xlabel("Date")
```

```
plt.ylabel("Trading Volume")
```

```
plt.xticks(rotation=45)
```

```
plt.show()
```

Output:



Result:

The trading volume of Alphabet Inc. stock between the specified dates was successfully filtered and displayed as a bar plot using Pandas and Matplotlib. The chart clearly represents the daily trading volume for the selected period.